

Estimate quotients using multiples

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Estimate and then calculate $7,392 \div 24$. Copy or complete the first step.

2. Estimate $5,985 \div 31$.

3. Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?

4. Miro says: Miro estimates division by rounding only the dividend. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Estimate and then calculate $7,392 \div 24$. Copy or complete the first step.
Answer About 300; exactly 308.
2. Estimate $5,985 \div 31$.
Answer About 200 because $6,000 \div 30$ is 200.
3. Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?
Answer 200.
4. Miro says: Miro estimates division by rounding only the dividend. Is this correct? Explain.
Answer Round both numbers to compatible values that are easy to divide.

Estimate quotients using multiples

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Estimate and then calculate $7,392 \div 24$.

6. Check this result using an inverse, estimate or known fact: Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?

2. Estimate $5,985 \div 31$.

7. Find a divisor between 20 and 30 that makes 6,048 divide exactly.

3. Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?

8. Identify and correct this misconception: Miro estimates division by rounding only the dividend.

4. Solve this independently and show every stage: Estimate and then calculate $7,392 \div 24$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Estimate $5,985 \div 31$.

10. Write and solve a similar question to: Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?

Teacher answers and checking prompts

1. Estimate and then calculate $7,392 \div 24$.
Answer About 300; exactly 308.
2. Estimate $5,985 \div 31$.
Answer About 200 because $6,000 \div 30$ is 200.
3. Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?
Answer 200.
4. Solve this independently and show every stage: Estimate and then calculate $7,392 \div 24$.
Answer About 300; exactly 308.
5. Use a second method or representation for: Estimate $5,985 \div 31$.
Answer Expected result: About 200 because $6,000 \div 30$ is 200. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?
Answer Expected result: 200. A valid check must be shown.
7. Find a divisor between 20 and 30 that makes 6,048 divide exactly.
Answer Possible answers include 21, 24 or 28.
8. Identify and correct this misconception: Miro estimates division by rounding only the dividend.
Answer Round both numbers to compatible values that are easy to divide.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Estimate quotients using multiples

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find a divisor between 20 and 30 that makes 6,048 divide exactly.

6. Create a new example that tests the same idea as: Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?

2. Prove the answer to this question, rather than only calculating it: Estimate and then calculate $7,392 \div 24$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Estimate $5,985 \div 31$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 200.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro estimates division by rounding only the dividend.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Find a divisor between 20 and 30 that makes 6,048 divide exactly.
Answer Possible answers include 21, 24 or 28.
2. Prove the answer to this question, rather than only calculating it: Estimate and then calculate $7,392 \div 24$.
Answer Expected result: About 300; exactly 308. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Estimate $5,985 \div 31$.
Answer Expected result: About 200 because $6,000 \div 30$ is 200. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 200.
Answer One valid reconstruction is: Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro estimates division by rounding only the dividend.
Answer Round both numbers to compatible values that are easy to divide.
6. Create a new example that tests the same idea as: Which is a better estimate for $8,316 \div 42$: 20, 200 or 2,000?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.